Dayeon Yoo

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Education

University of New Mexico, M.S. in Computer Engineering — Aug 2025 – May 2027
Quantum Information Science emphasis; GPA: 4.06/4.33
Advisor: Prof. Milad Marvian

University of New Mexico, Exchange Student in Computer Science; GPA: 4.2/4.33 — Aug 2024 – May 2025
Supported by the Korea-U.S. Student Exchange Program in STEM Fellowship

Dongguk University, B.S. in Multimedia Software Engineering; GPA: 4.33/4.5 — Mar 2020 – Aug 2025
Graduated with Highest Honors

Research Experience

Research Assistant, Center for Quantum Information and Control, University of New Mexico — Aug 2025 – Present

Manuscript in preparation: "Accelerating Quantum LDPC Code Optimization Through Adversarial Games" (with Prof. Milad Marvian).

- Developed an adversarial-game framework for quantum LDPC (qLDPC) code optimization, in which a code optimizer iteratively updates the code against an adversarial environment that identifies minimum-weight undetectable errors.
- Optimized hypergraph-product (HGP) codes across multiple benchmarks in seconds to minutes, including $[[1600, 64, 6]]$ to $[[1600, 64, 12]]$ in 34 seconds and $[[2500, 100, 10]]$ to $[[2500, 100, 14]]$ in 59 seconds; achieved $d = 9$ for $[[625, 25]]$ in 3 seconds, compared with $d = 8$ after 49 hours using a BP+OSD-based random-walk baseline.
- Optimized bivariate bicycle (BB) codes using low-weight logical operators, discovering codes including $[[112, 12, 8]]$, $[[228, 12, \leq 18]]$, and $[[340, 16, \leq 18]]$.

QU-REACH Research Intern, University of New Mexico — May 2025 – Aug 2025

- Developed graph-search methods for optimizing sparse structures in hypergraph-product quantum LDPC codes and benchmarked candidate-generation strategies against random-walk and simulated-annealing baselines.

Data Science Research Assistant, UNM Department of Marketing — Oct 2024 – May 2025

- Built a multimodal ML pipeline for 200K marketing advertisements, extracting image, video, audio, and text features using OpenCV, YOLOv8, Hugging Face Transformers, and speech-processing models for downstream analysis of creative performance and audience engagement.
- Engineered visual and audio features including object detection, facial emotion, motion, color, speech transcription, sentiment, and acoustic features; performed statistical analysis to identify relationships with audience engagement and brand perception.

Undergraduate Researcher, University of New Mexico — Aug 2024 – May 2025

- Implemented and evaluated a convolutional-autoencoder-based unsupervised anomaly-detection pipeline for experimental MUED diffraction images.
- Trained the CAE on 100 diffraction images and evaluated reconstruction-error-based anomaly detection across 1,521 experimental images for automated data-quality screening.

Research Manuscript

D. Yoo, M. Marvian, "Accelerating Quantum LDPC Code Optimization Through Adversarial Games," *manuscript in preparation*.

Presentations and Posters

Accelerating Quantum LDPC Code Optimization Through Adversarial Games — 2026
Poster presented at SQuInT 2026

Property Testing — 2026
Presentation, CQuIC Summer Course: Quantum Learning Theory

Fast and Accurate Neural-Network Decoders for Quantum Error-Correcting Codes — 2026
Presentation, UNM QEC Course

Barium-137 Trapped-Ion Quantum Computing — 2026

Presentation, UNM Platforms of Quantum Tech Course

Optimizing Hypergraph Product Codes 2025

Poster Presented at QU-REACH Poster

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, C, HTML/CSS, MATLAB

QEC: qLDPC codes (HGP, BB), BP+OSD, Stim, Qiskit, Monte Carlo simulation

ML & Data: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, pandas, NumPy, SciPy, OpenCV

Tools & Frameworks: HPC, Docker, PostgreSQL, HDF5, Slurm, Linux, Git, AWS, GCP, React Native, Next.js, NestJS

Work Experience

Co-Founder & Software Engineer, SnapIT! Jun 2024 – May 2025

- Launched an AI-powered iOS photo booth app on the App Store, reaching 2K+ downloads and reducing 4-cut photo generation time by 50% through algorithm and application-performance optimization; selected as a Top 10 finalist in the Epson Innovation Challenge.
- Deployed the app to the App Store and secured B2B partnerships for four local events, managing live deployments, stakeholder communication, and user feedback.

Software Engineer, Divers Nov 2022 – Feb 2024

- Led full-stack development of a cross-platform mobile app using React Native, Next.js, NestJS, PostgreSQL, and AWS, scaling the platform to 15K+ users.
- Diagnosed production crashes and API/network failures using application telemetry, optimized runtime and memory performance, and validated releases through automated, unit, and integration testing.

Selected Projects

NASA Space Apps Challenge – Fire Fighter (Website) *1st Place, 2024*

- Developed an interactive wildfire analytics platform using NASA Earth-observation APIs, implementing dynamic geospatial visualizations of wildfire activity and effects on air quality, biodiversity, and agriculture.

Microsoft GenAI Hackathon – RAG Q&A Bot (Website) *3rd Place, 2024*

- Developed a retrieval-augmented generation (RAG) system for government documents, combining document retrieval and generative AI to produce context-grounded responses.

Lobo Hackathon – LLM Legal Information Platform (Website) *Prize Winner, 2025*

- Developed an LLM-powered platform that processes U.S. government documents and generates accessible summaries to help nonprofit and public users understand complex legal information.

Honors, Awards & Fellowships

Korea-U.S. Student Exchange Program in STEM Fellowship 2024

Academic Merit Scholarship, Dongguk University 2021 – 2022

Awarded for ranking 1st in the major by GPA; received for three semesters.

Leadership & Extracurricular Activities

Member, IEEE UNM chapter – Represented IEEE at CES 2026 and CES 2025 2024 – Present

Member, Society of Women Engineers (SWE) UNM chapter – Attended WE24 and WE25 2024 – Present

Founder & President, Hack The World Software Engineering Club – University of New Mexico 2024 – 2025

Officer, Korean Language and Culture Club – University of New Mexico 2024 – 2025